

This document corresponds to Lab 6 of the myCobot laboratory sequence and addresses trajectory planning for smooth point-to-point robot motion. The practical work is carried out in Python using Jupyter notebooks within the myCobot 280 course environment, where trajectories are generated offline and then used in the experimental workflow.

The primary focus is joint-space trajectory planning under velocity and acceleration constraints. The motion profile consists of an acceleration phase, an optional constant-velocity cruise phase, and a deceleration phase. For short motions, the cruise phase vanishes and the result is a triangular profile rather than a full trapezoidal one. The lab therefore studies minimum-time point-to-point motion generation under kinematic limits and its practical use for the myCobot platform. The provided myCobot codebase contains a practical implementation of this profile generation in `_build_ee_profile()`, which computes the acceleration, cruise, and deceleration phases under velocity and acceleration limits and handles both the trapezoidal and triangular cases.

Prior to the lab, students should read this document carefully and complete the corresponding trajectory-planning quiz on Moodle.

Learning Objectives Trajectory Planning Lab

This lab develops the basic concepts required to construct constrained point-to-point trajectories for robot joints and to interpret their practical role in smooth robot motion.

By the end of the lab, students will be able to:

- explain the structure of trapezoidal velocity profiles for point-to-point motion,
- compute acceleration, cruise, and deceleration durations from motion and limit constraints,
- identify when the cruise phase disappears and the motion becomes triangular,
- generate feasible joint trajectories under prescribed velocity and acceleration limits,
- use these trajectories within the practical myCobot workflow implemented in Python and Jupyter notebooks.

Short Questions

- 1) What is the objective of trajectory planning?
- 2) What is the difference between a path and trajectory?
- 3) Which of the following statements about trajectory planning are true?
 - a) Trajectory planning considers maximum joint velocities and accelerations.
 - b) The acceleration profile of a planned trajectory has to be continuous.
 - c) Trajectory planning is based on a dynamic model of the robot.
 - d) Joint positions and velocities should be continuous functions of time.
- 4) Sketch the corresponding position and acceleration profile for the trapezoidal velocity profile below.

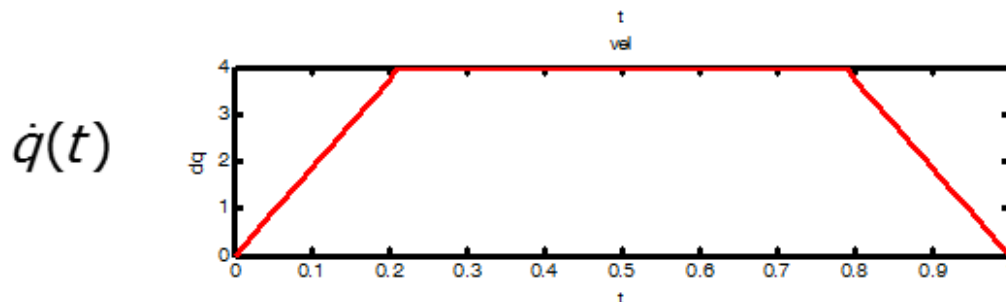


Figure 1: velocity profile

- 5) When does a point-to-point motion result in a triangular velocity profile instead of a full trapezoidal profile?

- 6) What are the three phases of a trapezoidal velocity profile, and what happens in each phase?

- 7) How do the maximum velocity and maximum acceleration influence the total motion time?

- 8) Why are smooth velocity profiles preferable to abrupt step changes in joint commands?

Trajectory Planning with Trapezoidal Velocity Profiles

Point-to-point robot motion should be generated smoothly and under prescribed velocity and acceleration constraints. In this lab, the relevant reference motion is constructed primarily in joint space. A standard profile consists of an acceleration phase, an optional constant-velocity cruise phase, and a deceleration phase. For short moves, the cruise phase disappears and the motion becomes triangular instead of fully trapezoidal.

Consider a scalar point-to-point motion from q_0 to q_f with displacement

$$\Delta q = q_f - q_0. \quad (1)$$

The profile is determined by the maximum velocity $v_{\max}$ and the maximum acceleration $a_{\max}$. For a full trapezoidal profile, the acceleration time is

$$t_a = \frac{v_{\max}}{a_{\max}}, \quad (2)$$

and the distance covered during acceleration and deceleration is

$$\Delta q_{\text{accdec}} = \frac{v_{\max}^2}{a_{\max}}. \quad (3)$$

If $|\Delta q| > \Delta q_{\text{accdec}}$, the motion is trapezoidal. If $|\Delta q| \leq \Delta q_{\text{accdec}}$, the motion is triangular.

For the trapezoidal case,

$$t_c = \frac{|\Delta q| - v_{\max}^2/a_{\max}}{v_{\max}}, \quad (4)$$

$$t_f = 2t_a + t_c. \quad (5)$$

For the triangular case,

$$t_a = \sqrt{\frac{|\Delta q|}{a_{\max}}}, \quad (6)$$

$$v_{\text{peak}} = a_{\max} t_a, \quad (7)$$

$$t_f = 2t_a. \quad (8)$$

The sign of Δq determines the direction of motion, while the phase durations are computed from $|\Delta q|$.

Motion Phases

During the acceleration phase, the joint speed increases linearly from rest to the cruise or peak value, and the position evolves quadratically in time. During the cruise phase,

the velocity remains constant and the position changes linearly. During the deceleration phase, the velocity decreases linearly back to zero and the position again follows a quadratic time law. Consequently, the velocity profile is piecewise linear, while the position profile is piecewise quadratic or linear-quadratic depending on whether a cruise phase is present.

Trajectory Planning for the myCobot 280

In the practical Jupyter notebook, these profiles are used to generate smooth joint-space reference motions for the myCobot 280. This avoids abrupt step commands and helps keep the commanded motion feasible with respect to joint velocity and acceleration limits. Task-space planning may later be considered as an extension, but the present lab focuses on joint-space trajectory generation.

Worked Example

Consider a single-joint motion from $q_0 = 0^\circ$ to $q_f = 90^\circ$ with $v_{\max} = 45^\circ/\text{s}$ and $a_{\max} = 90^\circ/\text{s}^2$. The displacement is $|\Delta q| = 90^\circ$. For a full trapezoidal profile,

$$t_a = \frac{45}{90} = 0.5 \text{ s}, \quad (9)$$

$$\Delta q_{\text{accdec}} = \frac{45^2}{90} = 22.5^\circ. \quad (10)$$

Since $90^\circ > 22.5^\circ$, the motion is trapezoidal. The cruise duration is

$$t_c = \frac{90 - 22.5}{45} = 1.5 \text{ s}, \quad (11)$$

and the total motion time is

$$t_f = 2 \cdot 0.5 + 1.5 = 2.5 \text{ s}. \quad (12)$$

This example illustrates how the move length, together with the velocity and acceleration limits, determines whether the cruise phase is present.

Table 1: Example single-joint move

q_0 [°]	q_f [°]	$v_{\max}$ [°/s]	$a_{\max}$ [°/s ²]	profile	t_a [s]	t_c [s]	t_f [s]
0	90	45	90	trapezoidal	0.5	1.5	2.5

Once a trajectory has been parameterized, it provides time-dependent position, velocity, and acceleration references that can be sampled in the notebook and used for smooth robot motion.

Competences

After completing this lab, students should be able to formulate constrained point-to-point joint motions, construct triangular and trapezoidal velocity profiles, interpret the effect of velocity and acceleration limits on the resulting motion, and use such profiles to obtain smooth robot motion in practice.